Worksheet 7

Integrated cases

Fully worked solutions

Badr ABOUFARASSE, PhD

2. Assessment and exam practice

Practise independently with assessment and final-exam style questions.

Assessment / exam practice 70 min

Case A – Continuous-assessment preparation

A school surveys 200 students about the maximum price they would accept for an optional service:

Price	$[5,10[$	$[10,15[$	$[15,20[$	$[20,30[$	$[30,40[$	$[40,60[$
n_i	10	25	45	60	40	20

Identify population, unit and variable type; build the complete table with widths, densities, frequencies, ICC, DCC and midpoints; draw histogram and cumulative curves; calculate modal class, median, quartiles, mean, population variance, standard deviation and CV; find the price affordable to about 65%; then classify a satisfied/not satisfied variable and choose a graph.

Case A – Detailed solution

Observed population: 200 surveyed students. Unit: one student. Variable: maximum acceptable price, continuous quantitative.

Class						
n_i	10	25	45	60	40	20
a_i	5	5	5	10	10	20
d_i	2	5	9	6	4	1
ICC	10	35	80	140	180	200
DCC	200	190	165	120	60	20

Class	$[5,10[$	$[10,15[$	$[15,20[$	$[20,30[$	$[30,40[$	$[40,60[$
c_i	7.5	12.5	17.5	25	35	50

The modal class is $[15,20[$.

$$Q_1 \approx 16.67, \qquad Me \approx 23.33, \qquad Q_3 = 32.50.$$

$$\bar{x} = \frac{5075}{200} = 25.375.$$

$$V = 129.859375, \qquad \sigma \approx 11.396, \qquad CV \approx 44.91\%.$$

For 65% to afford at least price p, use the 35th percentile:

$$0.35(200) = 70,$$

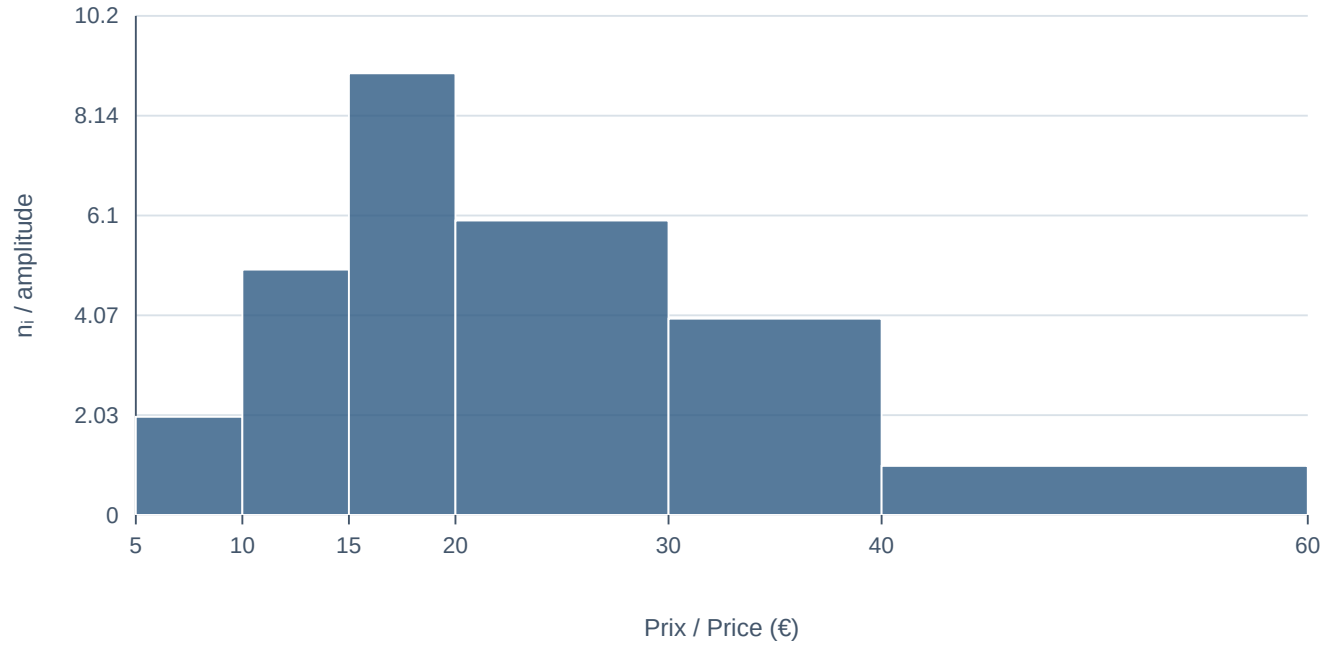
$$p = 15 + 5\frac{70-35}{45} \approx 18.89 \text{ euros}.$$

Satisfied/not satisfied is a binary nominal qualitative variable; a bar chart is appropriate.

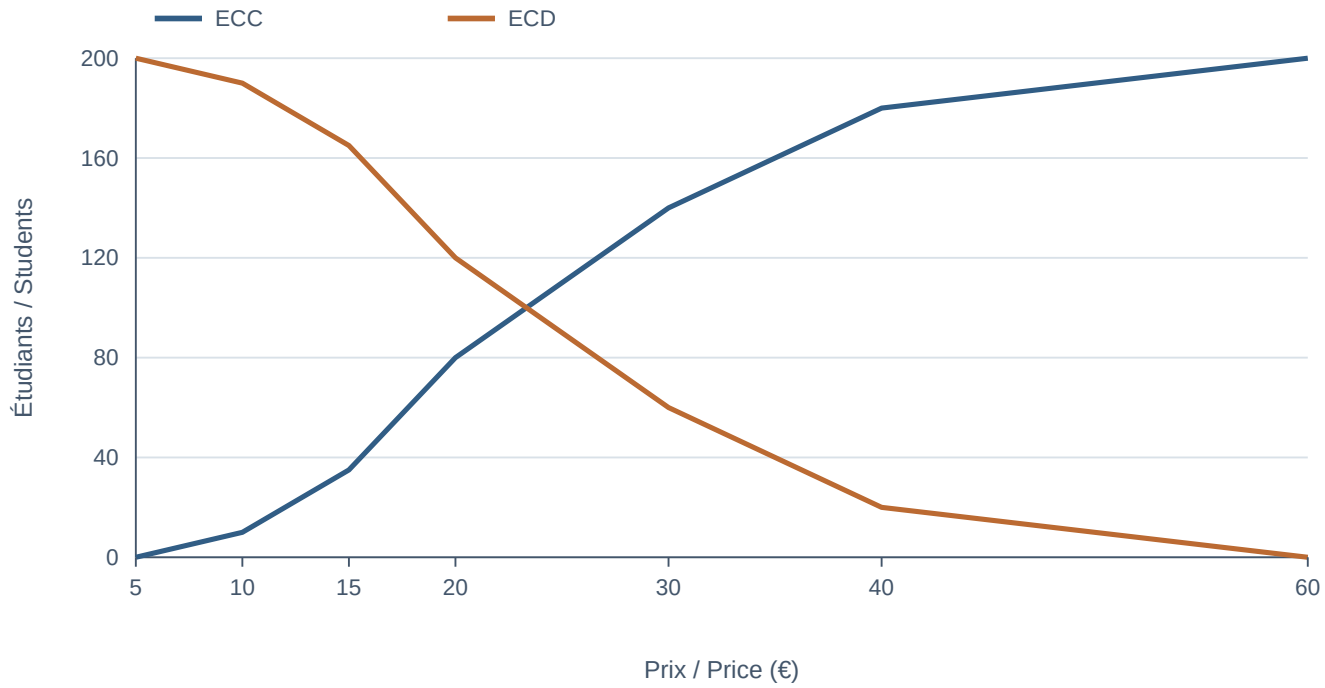
Full table, including all cumulative frequencies and the columns required for the mean and variance:

Classe	n_i	a_i	d_i	f_i (%)	ECC	ECD	FCC (%)	FCD (%)	c_i	$n_i c_i$	$n_i c_i^2$
[5,10[	10	5	2	5	10	200	5	100	7.5	75	562.5
[10,15[	25	5	5	12.5	35	190	17.5	95	12.5	312.5	3906.25
[15,20[	45	5	9	22.5	80	165	40	82.5	17.5	787.5	13781.25
[20,30[	60	10	6	30	140	120	70	60	25	1500	37500
[30,40[	40	10	4	20	180	60	90	30	35	1400	49000
[40,60[	20	20	1	10	200	20	100	10	50	1000	50000
TOTAL	200	—	—	100	200	—	100	—	—	5075	154750

Histogram heights are the densities: 2, 5, 9, 6, 4 and 1.



Cas A — histogramme des densités / Density histogram



Cas A — effectifs cumulés / Cumulative counts

$$\bar{x} = 5075/200 = 25.375, \quad V = 154750/200 - (25.375)^2 = 129.859375.$$

$$\sigma = \sqrt{129.859375} = 11.395586, \quad CV = 100(11.395586/25.375) = 44.9087\%.$$

$$Me = 20 + \frac{100-80}{60}(10) = 23.333333, \quad Q_1 = 15 + \frac{50-35}{45}(5) = 16.666667, \quad Q_3 = 30 + \frac{150-140}{40}(10) = 32.5.$$

$$P_{35} = 15 + \frac{70-35}{45}(5) = 18.888889 \text{ euros}.$$

The price affordable to approximately 65% of students is the 35th percentile, not the 65th. Results are approximate because the data are grouped.

Assessment / exam practice 110 min

Case B — New mock final exam — 110 minutes

Independent mock exam: the data below differ from exercises in chapters 4–6. Show every calculation step and justify rounding.

Part 1 — Two regressions (40 min)

t	1	2	3	4	5	6
x	5	6	8	9	11	13
y	12	15	18	20	24	29

1. Build t, x, y, t^2, x^2, y^2, tx, xy table; calculate both correlations and the regressions x on t and y on x. 2. Forecast x and y for t=8; qualify the forecast.

Part 2 — Quarterly series (40 min)

Année / Year	Q1	Q2	Q3	Q4
A	72	88	80	110
B	78	96	88	121

1. Plot the series; calculate MA4 and CMA4. 2. With the supplied trend $T_t=2.8t+78$, compute all eight ratios Y_t/T_t. 3. Average by quarter, normalise (sum 4) and forecast t=9 to 12.

Part 3 — Index numbers (30 min)

Produit / Product	p_0	q_0	p_1	q_1
A	10	40	12	45
B	8	60	9	55
C	20	20	22	25

Compute all four aggregates; Laspeyres, Paasche and Fisher price and quantity indices; the value index; and verify Fisher's factor reversal.

Case B — Fully worked mock exam solution

Part 1 — Regression

t	x	y	t^2	x^2	y^2	t×x	x×y
1	5	12	1	25	144	5	60
2	6	15	4	36	225	12	90
3	8	18	9	64	324	24	144
4	9	20	16	81	400	36	180

t	x	y	t^2	x^2	y^2	t×x	x×y
5	11	24	25	121	576	55	264
6	13	29	36	169	841	78	377
Σt=21	52	118	91	496	2510	210	1115

$$\bar{t} = 21/6 = 3.5, \quad \bar{x} = 52/6 = 8.666667, \quad \bar{y} = 118/6 = 19.666667.$$

$$V(t) = 91/6 - 3.5^2 = 2.916667, \quad V(x) = 496/6 - (52/6)^2 = 7.555556, \quad V(y) = 2510/6 - (118/6)^2 = 31.555556.$$

$$\mathrm{Cov}(t,x) = 210/6 - 3.5(52/6) = 4.666667, \quad r_{tx} = \frac{4.666667}{\sqrt{2.916667(7.555556)}} = .994100.$$

$$a_1 = 4.666667/2.916667 = 1.6, \quad b_1 = 52/6 - 1.6(3.5) = 3.066667, \quad \widehat{x}_t = 1.6t + 3.066667.$$

$$\mathrm{Cov}(x,y) = 1115/6 - (52/6)(118/6) = 15.388889, \quad r_{xy} = \frac{15.388889}{\sqrt{7.555556(31.555556)}} = .996635.$$

$$a_2 = 15.388889/7.555556 = 2.036765, \quad b_2 = 118/6 - a_2(52/6) = 2.014706.$$

$$\widehat{x}_8 = 1.6(8) + 3.066667 = 15.866667, \quad \hat{y}_8 = 2.036765(15.866667) + 2.014706 = 34.331373.$$

Period 8 is beyond the observed 1–6 range: extrapolation. Correlation does not imply causation.

Part 2 — Time series

$$MA_{2.5} = (72 + 88 + 80 + 110)/4 = 87.5, \quad MA_{3.5} = (88 + 80 + 110 + 78)/4 = 89.$$

Fenêtre / Window	MA4	Période médiane / Midperiod
1–4	87.5	2.5
2–5	89	3.5
3–6	91	4.5
4–7	93	5.5
5–8	95.75	6.5

$$CMA_3 = (87.5 + 89)/2 = 88.25, \quad CMA_4 = (89 + 91)/2 = 90.$$

t	CMA4
3	88.25
4	90
5	92
6	94.375

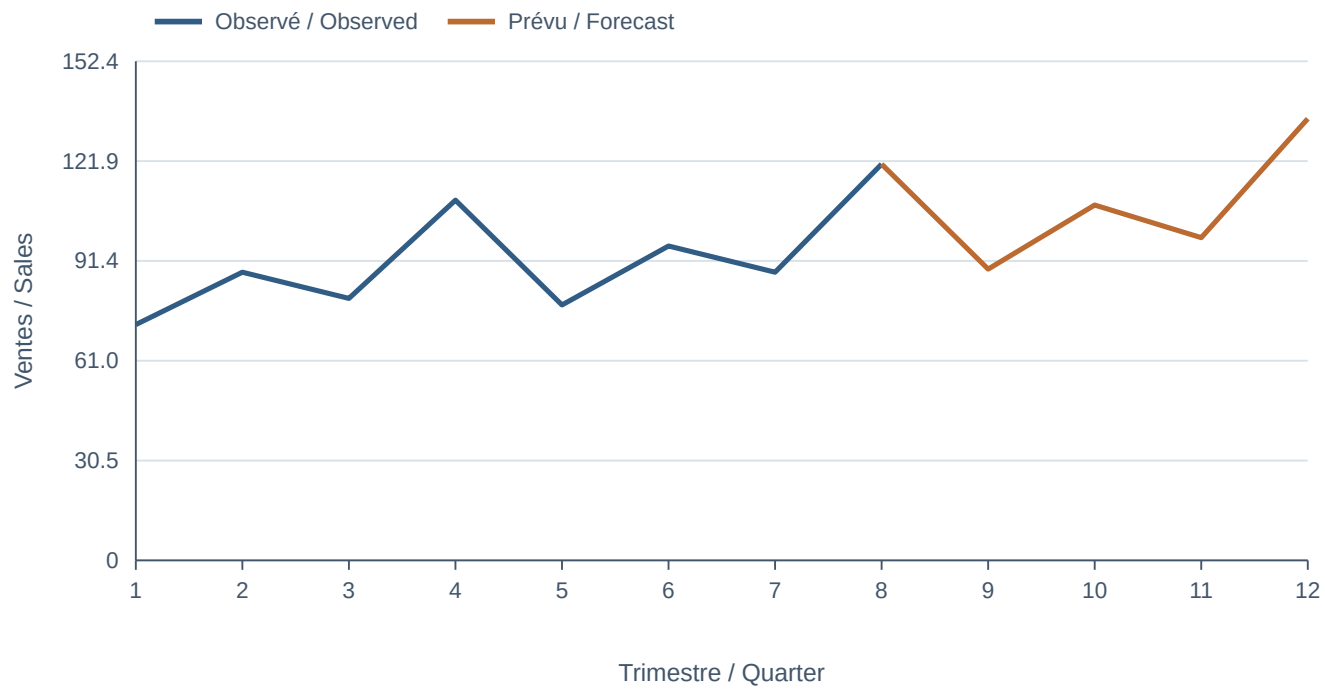
t	Y_t	T_t=2.8t+78	Y_t/T_t
1	72	80.8	0.891089
2	88	83.6	1.052632
3	80	86.4	0.925926
4	110	89.2	1.233184

t	Y_t	$T_t=2.8t+78$	Y_t/T_t
5	78	92	0.847826
6	96	94.8	1.012658
7	88	97.6	0.901639
8	121	100.4	1.205179

$$\bar{S}_q = (S_q + S_{q+4})/2, \qquad S'_q = \frac{4\bar{S}_q}{\sum_{k=1}^{4} \bar{S}_k}.$$

Quarter	Moyenne / Mean	S' normalisé / Normalised
Q1	0.869458	0.861902
Q2	1.032645	1.023671
Q3	0.913783	0.905841
Q4	1.219182	1.208586
TOTAL	4.035067	4

t	T_t	S'	Y prévu / Forecast
9	103.2	0.861902	88.9482
10	106	1.023671	108.5091
11	108.8	0.905841	98.5555
12	111.6	1.208586	134.8782



Cas B — série trimestrielle / Quarterly series

Forecasts t=9 to 12: 88.95, 108.51, 98.56 and 134.88.

Part 3 — Indices

Produit / Product	p_0q_0	p_1q_0	p_0q_1	p_1q_1
A	400	480	450	540
B	480	540	440	495
C	400	440	500	550
Σ	1280	1460	1390	1585

$$L_P = 100(1460/1280) = 114.0625, \quad P_P = 100(1585/1390) = 114.028777.$$

$$L_Q = 100(1390/1280) = 108.59375, \quad P_Q = 100(1585/1460) = 108.561644.$$

$$F_P = \sqrt{L_P P_P} = 114.045637, \quad F_Q = \sqrt{L_Q P_Q} = 108.577696.$$

$$I_V = 100(1585/1280) = 123.828125, \quad F_P F_Q/100 = 123.828125.$$

Total value rises by 23.83%, reflecting both price and quantity changes.